

Domain / Problem Area:

The project focuses on predicting an album's genre based on descriptive metadata. This is relevant to music streaming platforms and recommendation systems, since large libraries of songs and albums need to be categorized and suggested to users.

Approach & Methods:

I used three supervised classification models from scikit-learn: Logistic Regression, Decision Tree Classifier, and a k-Nearest Neighbors (KNN) Classifier. I also used GridSearchCV with 5-fold cross-validation to tune hyperparameters for all models and identify the best-performing configurations.

Tools & Technologies:

The project was implemented in Python using a Jupyter Notebook. I used pandas and NumPy for data processing, scikit-learn for machine learning models and preprocessing, and Matplotlib and Seaborn for visualization. The dataset consisted of album genres and descriptor terms, which were transformed into numerical features using CountVectorizer for use in classification.

Results:

Logistic Regression performed the best, achieving an accuracy of approximately 50%. This indicates that album descriptors contain useful information for predicting genre, although performance is limited by class imbalance and the relatively small number of samples per genre. Some genres, such as Singer-Songwriter and Conscious Hip Hop, were predicted more accurately due to more distinctive descriptor patterns, while others were more difficult to classify due to overlapping features.